

PRACTICE EXAMPLES

1) Create Table

```
[mysql> CREATE DATABASE student_db;
Query OK, 1 row affected (0.005 sec)

[mysql> USE student_db;
Database changed
mysql> CREATE TABLE student_marks1 (
    ->     roll_no INT PRIMARY KEY,
    ->     name VARCHAR(50),
    ->     subject VARCHAR(50),
    ->     marks DECIMAL(5,2)
    -> );
Query OK, 0 rows affected (0.018 sec)

[mysql> SHOW TABLES;
+-----+
| Tables_in_student_db |
+-----+
| student_marks1       |
+-----+
1 row in set (0.001 sec)
```

2) Insert Data

```
mysql> INSERT INTO student_marks1 (roll_no, name, subject, marks) VALUES
    -> (1, 'Ravi', 'Math', 85.50),
    -> (2, 'Sita', 'Math', 92.75),
    -> (3, 'Anil', 'Math', 78.40),
    -> (4, 'Priya', 'Math', 88.90),
    -> (5, 'Vijay', 'Math', 80.25),
    -> (6, 'subbusir', 'aws', 98.50),
    -> (7, 'dwaraka', 'dbms', 95.75),
    -> (8, 'ranjani', 'english', 97.40),
    -> (9, 'kavithamam', 'aws1', 99.90),
    -> (10, 'seetha', 'azure', 82.25);
Query OK, 10 rows affected (0.004 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

3) Simple Select

```
[mysql> SELECT * FROM student_marks1;
```

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
3	Anil	Math	78.40
4	Priya	Math	88.90
5	Vijay	Math	80.25
6	subbusir	aws	98.50
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
9	kavithamam	aws1	99.90
10	seetha	azure	82.25

```
10 rows in set (0.000 sec)
```

4) Aggregate Functions

```
mysql> SELECT COUNT(*) AS total_students
-> FROM student_marks1;
```

total_students
10

```
1 row in set (0.004 sec)
```



```
mysql> SELECT SUM(marks)
-> FROM student_marks1;
```

SUM(marks)
899.60

```
1 row in set (0.001 sec)
```

```
mysql> SELECT AVG(marks)
[   -> FROM student_marks1;
+-----+
| AVG(marks) |
+-----+
| 89.960000 |
+-----+
1 row in set (0.000 sec)

mysql> SELECT MAX(marks)
[   -> FROM student_marks1;
+-----+
| MAX(marks) |
+-----+
| 99.90 |
+-----+
1 row in set (0.002 sec)

mysql> SELECT MIN(marks)
[   -> FROM student_marks1;
+-----+
| MIN(marks) |
+-----+
| 78.40 |
+-----+
1 row in set (0.000 sec)
```

5) Filtering with WHERE

```
mysql> SELECT *
[   -> FROM student_marks1
[   -> WHERE marks > 85;
+-----+-----+-----+-----+
| roll_no | name      | subject | marks |
+-----+-----+-----+-----+
| 1       | Ravi      | Math    | 85.50 |
| 2       | Sita      | Math    | 92.75 |
| 4       | Priya     | Math    | 88.90 |
| 6       | subbusir  | aws     | 98.50 |
| 7       | dwaraka   | dbms    | 95.75 |
| 8       | ranjani   | english | 97.40 |
| 9       | kavithamam | aws1    | 99.90 |
+-----+-----+-----+-----+
7 rows in set (0.001 sec)

mysql> SELECT *
[   -> FROM student_marks1
[   -> WHERE marks >= 90;
+-----+-----+-----+-----+
| roll_no | name      | subject | marks |
+-----+-----+-----+-----+
| 2       | Sita      | Math    | 92.75 |
| 6       | subbusir  | aws     | 98.50 |
| 7       | dwaraka   | dbms    | 95.75 |
| 8       | ranjani   | english | 97.40 |
| 9       | kavithamam | aws1    | 99.90 |
+-----+-----+-----+-----+
5 rows in set (0.000 sec)
```

```
mysql> SELECT *
      -> FROM student_marks1
      -> WHERE marks < 80;
```

roll_no	name	subject	marks
3	Anil	Math	78.40

1 row in set (0.000 sec)

```
mysql> SELECT *
      -> FROM student_marks1
      -> WHERE marks BETWEEN 80 AND 90;
```

roll_no	name	subject	marks
1	Ravi	Math	85.50
4	Priya	Math	88.90
5	Vijay	Math	80.25
10	seetha	azure	82.25

4 rows in set (0.001 sec)

```
mysql> SELECT *
      -> FROM student_marks1
      -> WHERE name LIKE 'P%';
```

roll_no	name	subject	marks
4	Priya	Math	88.90

1 row in set (0.003 sec)

```
mysql> SELECT *
      -> FROM student_marks1
      -> WHERE name IN ('Ravi', 'Sita', 'Vijay');
```

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
5	Vijay	Math	80.25

3 rows in set (0.001 sec)

```
mysql> SELECT *
      -> FROM student_marks1
      -> WHERE marks > 85
      -> AND (subject = 'Math' OR name LIKE 'P%');
```

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
4	Priya	Math	88.90

3 rows in set (0.001 sec)

6) Update Queries

```
mysql> UPDATE student_marks1  
      -> SET marks = 90.00  
[      -> WHERE roll_no = 3;  
Query OK, 1 row affected (0.002 sec)  
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> UPDATE student_marks1  
      -> SET subject = 'Remedial Math'  
[      -> WHERE marks < 80;  
Query OK, 0 rows affected (0.001 sec)  
Rows matched: 0  Changed: 0  Warnings: 0
```

```
[mysql> SELECT * FROM student_marks1;
```

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
3	Anil	Math	90.00
4	Priya	Math	88.90
5	Vijay	Math	80.25
6	subbusir	aws	98.50
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
9	kavithamam	aws1	99.90
10	seetha	azure	82.25

```
10 rows in set (0.000 sec)
```

7) Delete Queries

```
mysql> DELETE FROM student_marks1  
[ -> WHERE roll_no = 5;  
Query OK, 1 row affected (0.002 sec)
```

```
mysql> DELETE FROM student_marks1  
[ -> WHERE marks < 75;  
Query OK, 0 rows affected (0.001 sec)
```

```
[mysql> SELECT * FROM student_marks1;
```

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
3	Anil	Math	90.00
4	Priya	Math	88.90
6	subbusir	aws	98.50
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
9	kavithamam	aws1	99.90
10	seetha	azure	82.25

```
9 rows in set (0.001 sec)
```

8) Sorting

```
mysql> SELECT * FROM student_marks1  
[ -> ORDER BY marks ASC;
```

roll_no	name	subject	marks
10	seetha	azure	82.25
1	Ravi	Math	85.50
4	Priya	Math	88.90
3	Anil	Math	90.00
2	Sita	Math	92.75
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
6	subbusir	aws	98.50
9	kavithamam	aws1	99.90

```
9 rows in set (0.001 sec)
```

```
mysql> SELECT * FROM student_marks1  
[ -> ORDER BY marks DESC;
```

roll_no	name	subject	marks
9	kavithamam	aws1	99.90
6	subbusir	aws	98.50
8	ranjani	english	97.40
7	dwaraka	dbms	95.75
2	Sita	Math	92.75
3	Anil	Math	90.00
4	Priya	Math	88.90
1	Ravi	Math	85.50
10	seetha	azure	82.25

9 rows in set (0.001 sec)

```
mysql> SELECT * FROM student_marks1  
[ -> ORDER BY name ASC;
```

roll_no	name	subject	marks
3	Anil	Math	90.00
7	dwaraka	dbms	95.75
9	kavithamam	aws1	99.90
4	Priya	Math	88.90
8	ranjani	english	97.40
1	Ravi	Math	85.50
10	seetha	azure	82.25
2	Sita	Math	92.75
6	subbusir	aws	98.50

9 rows in set (0.001 sec)

```
mysql> SELECT * FROM student_marks1  
[ -> ORDER BY name DESC;
```

roll_no	name	subject	marks
6	subbusir	aws	98.50
2	Sita	Math	92.75
10	seetha	azure	82.25
1	Ravi	Math	85.50
8	ranjani	english	97.40
4	Priya	Math	88.90
9	kavithamam	aws1	99.90
7	dwaraka	dbms	95.75
3	Anil	Math	90.00

9 rows in set (0.000 sec)

9) Group By

```
mysql> SELECT subject, SUM(marks) AS total_marks  
-> FROM student_marks1  
[ -> GROUP BY subject;
```

subject	total_marks
Math	357.15
aws	98.50
dbms	95.75
english	97.40
aws1	99.90
azure	82.25

6 rows in set (0.002 sec)

```
mysql> SELECT subject, AVG(marks) AS avg_marks  
-> FROM student_marks1  
[ -> GROUP BY subject;
```

subject	avg_marks
Math	89.287500
aws	98.500000
dbms	95.750000
english	97.400000
aws1	99.900000
azure	82.250000

6 rows in set (0.001 sec)

```
mysql> SELECT subject, COUNT(*) AS num_students  
-> FROM student_marks1  
[ -> GROUP BY subject;
```

subject	num_students
Math	4
aws	1
dbms	1
english	1
aws1	1
azure	1

6 rows in set (0.001 sec)

10) Having

```
mysql> SELECT subject, AVG(marks) AS avg_marks
      -> FROM student_marks1
      -> GROUP BY subject
      -> HAVING AVG(marks) > 90;
+-----+-----+
| subject | avg_marks |
+-----+-----+
| aws     | 98.500000 |
| dbms    | 95.750000 |
| english | 97.400000 |
| aws1    | 99.900000 |
+-----+-----+
4 rows in set (0.001 sec)
```

```
mysql> SELECT subject, COUNT(*) AS num_students
      -> FROM student_marks1
      -> GROUP BY subject
      -> HAVING COUNT(*) > 1;
+-----+-----+
| subject | num_students |
+-----+-----+
| Math    | 4            |
+-----+-----+
1 row in set (0.001 sec)
```

```
mysql> █
```